

# Grade 11 Physics — Practice Exam Answer Key

Thai Mattayom 5 (OBEC Curriculum) - 40 Multiple Choice Questions

## Quick Reference

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1  | D | 2  | C | 3  | C | 4  | C | 5  | B |
| 6  | C | 7  | B | 8  | B | 9  | C | 10 | C |
| 11 | B | 12 | B | 13 | B | 14 | C | 15 | B |
| 16 | B | 17 | B | 18 | B | 19 | A | 20 | D |
| 21 | C | 22 | B | 23 | B | 24 | B | 25 | B |
| 26 | A | 27 | B | 28 | B | 29 | A | 30 | B |
| 31 | A | 32 | A | 33 | B | 34 | B | 35 | B |
| 36 | C | 37 | B | 38 | C | 39 | B | 40 | A |

## Detailed Explanations

### Work, Energy, and Power

1. A person pushes a box with a force of 50 N through a displacement of 4 m in the direction of the force. How much work is done?

**Correct answer: (D) 200 J**

Explanation:  $W = F \cdot d = 50 \text{ N} \times 4 \text{ m} = 200 \text{ J}$  (force and displacement are in the same direction, so  $\cos\theta = 1$ ).

2. A force of 20 N acts at an angle of  $60^\circ$  to the direction of a 5 m displacement. What is the work done?

**Correct answer: (C) 50 J**

Explanation:  $W = F d \cos\theta = 20 \times 5 \times \cos 60^\circ = 100 \times 0.5 = 50 \text{ J}$ .

3. A waiter carries a tray horizontally at constant velocity across a room. What is the work done by the upward force from the waiter's hand on the tray?

**Correct answer: (C) Zero**

Explanation: The supporting force is vertical while the displacement is horizontal, so  $\theta = 90^\circ$  and  $\cos 90^\circ = 0$ , giving zero work.

4. A 2 kg object is lifted at constant velocity through a height of 3 m. Find the work done against gravity. ( $g = 10 \text{ m/s}^2$ )

**Correct answer: (C) 60 J**

Explanation:  $W = mgh = 2 \times 10 \times 3 = 60 \text{ J}$ .

5. Power is best defined as:

**Correct answer: (B) Work done  $\div$  time**

Explanation: Power is the rate at which work is done,  $P = W / t$ .

6. A motor lifts a 500 N load to a height of 10 m in 5 seconds. What is the power output of the motor (ignoring losses)?

**Correct answer: (C) 1000 W**

Explanation:  $P = W / t = (F \times d) / t = (500 \times 10) / 5 = 1000 \text{ W}$ .

## Momentum and Collisions

7. Momentum is defined as the product of:

**Correct answer: (B) Mass and velocity**

Explanation: Momentum  $p = m v$ , the product of an object's mass and its velocity.

8. A 1000 kg car traveling at 20 m/s collides with an identical stationary 1000 kg car, and the two lock together. Find their common velocity after the collision.

**Correct answer: (B) 10 m/s**

Explanation: Conservation of momentum:  $(1000)(20) = (2000)(v) \rightarrow v = 20000/2000 = 10 \text{ m/s}$ .

9. In a perfectly elastic collision between two objects:

**Correct answer: (C) Both momentum and kinetic energy are conserved**

Explanation: By definition, an elastic collision conserves both total momentum and total kinetic energy.

10. A 0.5 kg ball moving at 6 m/s hits a wall and bounces straight back with the same speed. What is the magnitude of the change in momentum?

**Correct answer: (C) 6 kg·m/s**

Explanation:  $\Delta p = m(v_f - v_i) = 0.5 \times (-6 - 6) = -6 \text{ kg·m/s}$ ; magnitude = 6 kg·m/s.

11. Impulse delivered to an object is equal to its:

**Correct answer: (B) Change in momentum**

Explanation: Impulse  $J = F\Delta t = \Delta p$ , the change in momentum.

12. A 0.02 kg bullet is fired from a 5 kg gun initially at rest. If the bullet leaves with a velocity of 400 m/s, what is the recoil speed of the gun?

**Correct answer: (B) 1.6 m/s**

Explanation: Momentum conservation:  $0 = (0.02)(400) + (5)(v_{\text{gun}}) \rightarrow v_{\text{gun}} = -8/5 = -1.6 \text{ m/s}$ , so the recoil speed is 1.6 m/s.

## Rotational Motion and Equilibrium

13. The torque produced by a force  $F$  applied at a perpendicular distance  $r$  from a pivot is given by:

**Correct answer: (B)  $\tau = F \times r$**

Explanation: Torque is the product of the applied force and the perpendicular distance from the pivot:  $\tau = F \times r$ .

14. Which condition must be satisfied for a rigid body to be in complete mechanical equilibrium?

**Correct answer: (C) Both net force = 0 and net torque = 0**

Explanation: Complete equilibrium requires both translational equilibrium ( $\Sigma F = 0$ ) and rotational equilibrium ( $\Sigma \tau = 0$ ).

15. A wrench is used to turn a bolt. Applying the same force but at a longer distance from the bolt will:

**Correct answer: (B) Increase the torque**

Explanation: Since  $\tau = F \times r$ , increasing  $r$  while  $F$  stays the same increases the torque.

16. The moment of inertia of an object depends on:

**Correct answer: (B) Mass and how it is distributed relative to the axis of rotation**

Explanation: Moment of inertia depends on both the mass of the object and how that mass is distributed relative to the rotation axis.

17. A force of 15 N is applied perpendicular to a wrench at a distance of 0.2 m from the bolt. What torque is produced?

Correct answer: (B) 3 N·m

Explanation:  $\tau = F \times r = 15 \times 0.2 = 3 \text{ N}\cdot\text{m}$ .

18. Two children sit on a seesaw. A 30 kg child sits 2 m from the pivot. How far from the pivot must a 40 kg child sit on the other side for balance?

Correct answer: (B) 1.5 m

Explanation: For balance, torques must be equal:  $30 \times 2 = 40 \times x \rightarrow x = 60/40 = 1.5 \text{ m}$ .

## Gravitation

19. According to Newton's law of universal gravitation, the gravitational force between two masses is:

Correct answer: (A) Directly proportional to the product of the masses and inversely proportional to the square of the distance between them

Explanation:  $F = G m_1 m_2 / r^2$  — force increases with the product of the masses and decreases with the square of the separation.

20. If the distance between two objects is doubled, the gravitational force between them becomes:

Correct answer: (D) One-quarter as strong

Explanation: Since  $F \propto 1/r^2$ , doubling  $r$  reduces the force to  $1/2^2 = 1/4$  of its original value.

21. The acceleration due to gravity on the surface of a planet depends on:

Correct answer: (C) Both the mass and radius of the planet

Explanation:  $g = GM/R^2$ , so it depends on both the planet's mass  $M$  and its radius  $R$ .

22. A satellite orbits Earth in a circular path. The gravitational force acting on it serves as:

Correct answer: (B) The centripetal force keeping it in orbit

Explanation: Gravity supplies the centripetal force required to keep the satellite moving along its circular orbit.

23. Weightlessness experienced by astronauts in an orbiting spacecraft occurs because:

Correct answer: (B) The spacecraft and astronauts are in continuous free fall together

Explanation: Gravity is still present in orbit; astronauts feel weightless because they and the spacecraft are falling toward Earth at the same rate.

24. If the mass of Earth were to double while its radius stayed the same, the gravitational acceleration at its surface would:

Correct answer: (B) Double

Explanation:  $g = GM/R^2$ ; doubling  $M$  while  $R$  is unchanged doubles  $g$ .

## Periodic Motion (SHM)

25. Simple harmonic motion (SHM) is characterized by a restoring force that is:

Correct answer: (B) Directly proportional to displacement and directed toward the equilibrium position

Explanation: SHM arises when the restoring force follows  $F = -kx$ , proportional to displacement and directed toward equilibrium.

26. At the equilibrium position of an object undergoing SHM:

Correct answer: (A) Velocity is maximum and acceleration is zero

Explanation: At equilibrium, displacement  $x = 0$ , so the restoring force (and acceleration) is zero, while speed is at its maximum.

27. The period of a simple pendulum depends on:

**Correct answer: (B) The length of the string and gravitational acceleration**

Explanation:  $T = 2\pi\sqrt{L/g}$ ; period depends only on the string length  $L$  and gravitational acceleration  $g$  (for small angles).

28. A simple pendulum has a length of 1 m. Using  $g = 10 \text{ m/s}^2$  and  $\pi^2 \approx 10$ , find its period.

**Correct answer: (B) 2 s**

Explanation:  $T = 2\pi\sqrt{L/g}$ ;  $T^2 = 4\pi^2 L/g = 4(10)(1)/10 = 4$ , so  $T = 2 \text{ s}$ .

29. If the length of a simple pendulum is quadrupled, its period will:

**Correct answer: (A) Double**

Explanation:  $T \propto \sqrt{L}$ , so quadrupling  $L$  increases  $T$  by a factor of  $\sqrt{4} = 2$ .

30. For a mass–spring system in SHM, increasing the spring constant  $k$  (keeping mass constant) will:

**Correct answer: (B) Decrease the period**

Explanation:  $T = 2\pi\sqrt{m/k}$ ; increasing  $k$  decreases the period.

31. The amplitude of an oscillating system in SHM represents:

**Correct answer: (A) The maximum displacement from equilibrium**

Explanation: Amplitude is defined as the maximum displacement from the equilibrium position.

32. A mass on a spring completes 10 oscillations in 5 seconds. What are its period and frequency?

**Correct answer: (A)  $T = 0.5 \text{ s}$ ,  $f = 2 \text{ Hz}$**

Explanation:  $T = \text{total time} / \text{number of oscillations} = 5/10 = 0.5 \text{ s}$ ;  $f = 1/T = 2 \text{ Hz}$ .

## Mechanical Waves and Sound

33. In a transverse wave, the particles of the medium vibrate:

**Correct answer: (B) Perpendicular to the direction of wave propagation**

Explanation: In a transverse wave, particle motion is perpendicular to the direction the wave travels.

34. Sound waves traveling through air are classified as:

**Correct answer: (B) Longitudinal waves**

Explanation: Sound waves are longitudinal — particles vibrate parallel to the direction of wave travel.

35. The speed of a wave is related to its wavelength ( $\lambda$ ) and frequency ( $f$ ) by the equation:

**Correct answer: (B)  $v = \lambda f$**

Explanation: Wave speed  $v = \lambda f$ , the product of wavelength and frequency.

36. A wave has a frequency of 250 Hz and a wavelength of 1.4 m. What is its speed?

**Correct answer: (C) 350 m/s**

Explanation:  $v = \lambda f = 1.4 \times 250 = 350 \text{ m/s}$ .

37. The pitch of a sound is most closely related to its:

**Correct answer: (B) Frequency**

Explanation: Pitch is perceived based on frequency — higher frequency sounds are perceived as higher pitched.

38. As a sound wave travels from air into water, which property remains unchanged?

**Correct answer: (C) Frequency**

Explanation: Frequency is set by the source and stays constant as a wave crosses into a new medium; speed and wavelength both change.

39. The loudness of a sound is primarily determined by the wave's:

Correct answer: (B) Amplitude

Explanation: Loudness (intensity) depends on the amplitude of the sound wave — larger amplitude means louder sound.

40. Two sound waves interfere constructively when they meet:

Correct answer: (A) In phase, with crest aligning with crest

Explanation: Constructive interference occurs when waves are in phase, so crests align with crests (and troughs with troughs), reinforcing the wave.